Ten years later, Einstein gave the world his answer with his special theory of relativity. There has been much debate regarding the intellectual roots of Einstein's discovery, but there is no doubt that his unshakable belief in simplicity played a critical role. Einstein was aware of at least some experiments that had failed to detect evidence for the existence of the aether.[2] So why dance around trying to find fault with the experiments? Instead, Einstein declared, take the simple approach: The experiments were failing to find the aether because there is no aether. And since Maxwell's equations describing the motion of light—the motion of electromagnetic waves—do not invoke any such medium, both experiment and theory would converge on the same conclusion: light, unlike any other kind of wave ever encountered, does not need a medium to carry it along. Light is a lone traveler. Light can travel through empty space.

But what, then, are we to make of Maxwell's equation giving light a speed of 670 million miles per hour? If there is no aether to provide the standard of rest, what is the *what* with respect to which this speed is to be interpreted? Again, Einstein bucked convention and answered with ultimate simplicity. If Maxwell's theory does not invoke any particular standard of rest, the most direct interpretation is that we don't need one. *The speed of light,* Einstein declared, *is 670 million miles per hour relative to anything and everything.*

Well, this is certainly a simple statement; it fit well a maxim often attributed to Einstein: "Make everything as simple as possible, but no simpler." The problem is that it also seems crazy. If you run after a departing beam of light, common sense dictates that from your perspective the speed of the departing light has to be less than 670 million miles per hour. If you run toward an approaching beam of light, common sense dictates that from your perspective the speed of the approaching light will be greater than 670 million miles per hour. Throughout his life, Einstein challenged common sense, and this time was no exception. He forcefully argued that regardless of how fast you move toward or away from a beam of light, you will always measure its speed to be 670 million miles per hour—not a bit faster, not a bit slower, no matter what. This would certainly solve the paradox that stumped him as a teenager: Maxwell's theory does not allow for stationary light because light *never* is stationary; regardless of your state of motion,